

Fisher Information Calculator

Multi-channel optimal reconstruction of $G(t)$ via Cramér–Rao bound

Model

```
Y_i(t) = Y0_i · (1 + c_i · G(t)) + noise_i    noise ~ N(0, σ²_i)
G(t)    = A_base + A_amp · cos(2πt)

Fisher Information:  F = Σ_i [ Y0_i² · c_i² / σ_i² ]
Cramér–Rao bound:   σ²_G ≥ 1 / F
Optimal estimate:    G^(t) = [Σ_i (c_i·Y0_i/σ²_i)(Y_i(t)-Y0_i)] / F
```

Scalar Field Parameters

A_BASE

A_AMP

Measurement Streams

Add independent measurement channels. Each stream $Y_i(t) = Y0_i \cdot (1 + c_i \cdot G(t)) + \epsilon_i$. More streams → tighter Cramér–Rao bound.

STREAM	Y0 _i	c _i	Σ _i
Stream 1	1	1	0.0005
Stream 2	1	0.5	0.0003

STREAM	Y_{0i}	C_i	Σ_i
Stream 3	1	0.3	0.0004

+ Add Stream

- Remove

Compute

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